MULTI-CHAMBER EXPANSION AND REGIONAL COMPLIANCE MAPPING EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device may incorporate multiple expandable regions configured to measure localized mechanical properties along different portions of an intraluminal structure. Multi-chamber expansion architectures enable regional compliance mapping rather than single-point measurement.

The insertable section may include two or more independently inflatable chambers positioned longitudinally, circumferentially, or in segmented arrangements. Independent control of each chamber allows measurement of localized deformation characteristics and spatial variation in tissue response.

In certain embodiments, chambers may be inflated sequentially, simultaneously, or according to programmable expansion patterns. Sequential inflation may allow reconstruction of compliance profiles along the insertion axis, while simultaneous inflation may evaluate distributed mechanical interaction.

Regional compliance mapping may derive parameters including stiffness gradients, asymmetry characteristics, deformation uniformity, and localized resistance variations. Such parameters may provide enhanced physiologic characterization compared to single-chamber expansion systems.

In some implementations, pressure sensors or strain sensing elements may be distributed across multiple chambers to capture independent expansion responses. Sensor fusion algorithms may integrate chamber-specific measurements into composite compliance models.

Multi-chamber embodiments may further enable adaptive measurement strategies in which subsequent chambers adjust inflation behavior based on measurements obtained from preceding chambers. Adaptive sequencing may improve measurement efficiency and user comfort.

The device may also implement differential inflation modes in which opposing chambers expand at different pressures to detect asymmetry or directional compliance differences within the measured region.

In certain configurations, regional compliance data may be translated into graphical maps, indexed scores, or abstracted structural profiles within associated computational systems. Mapping outputs may support longitudinal tracking or compatibility modeling applications.

Multi-chamber systems may include shared or independent fluid pathways, valves, and sensors, and may be implemented using modular inflatable structures allowing simplified manufacturing or upgradeability.

Accordingly, multi-chamber expansion embodiments extend the functionality of the disclosed device from single-region measurement toward distributed mechanical mapping, enabling richer physiologic analysis and supporting advanced computational modeling frameworks.